

High-density Evanescent Chip Coupling with Detachable Fiber Connector for Co-packaged Optics

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Abstract: A glass interconnect for fiber-to-chip coupling with integrated pitch conversion and detachable fiber array connector for CPO is reported. The minimum evanescent coupling loss is 0.38 dB at 1337 nm for flip-chip assembled SiN chip.

1. Introduction

As the world undergoes a digital transformation, the need for faster data transmission and higher bandwidth intensifies, especially given the rapid growth driven by artificial intelligence (AI) computer clusters [1]. Co-packaged optics (CPO) is a promising technology that integrates optical components with electronic chips, providing a high bandwidth, low latency, and power-efficient solution for data transmission [2]. Silicon Photonic Integrated Circuits (PICs) play a crucial role in advancing CPO due to their compatibility with existing CMOS fabrication processes, which allows for the integration of optical and electronic components on a single chip. This results in lower costs and promotes high-volume manufacturing and wafer-level packaging [3]. However, a critical aspect of optical communication systems is the fiber-to-chip coupling which serves as a bridge between the fiber cable network and the electronic component packaging. Various fiber-to-chip coupling solutions exist, each with its advantages and drawbacks. These include edge coupling, tapered waveguides, integrated mode-expansion, grating coupling, and evanescent or adiabatic coupling [4].

A key requirement for CPO solution is the need for advanced methods of high-density optical chip interconnects and detachable optical fiber connectivity as shown in Figure 1A to enable high-volume and low-cost electronic packaging. The performance of waveguide materials, particularly in terms of optical loss at specific wavelengths, is crucial in the realm of optical communication systems. Different waveguide platforms such as polymer, glass, and silicon nitride (SiN) integrated waveguides are considered based on their performance in the C- and O-band wavelength range. This work will delve into a glass interconnect transfer waveguide solution as an optical-mechanical interface between the photonic integrated circuit and the fiber cable with an assembled CPO connector (Figure 1B). The glass interconnect supports high-density coupling via evanescent couplers to the PIC and detachable fiber edge-coupling to the fiber cable. Longer term, a single glass substrate with integrated glass interconnects, as shown in Figure 1C, could be combined with an electrical flip-chip assembly of PIC transceivers for CPO with Application Specific Integrated Circuit (ASIC).

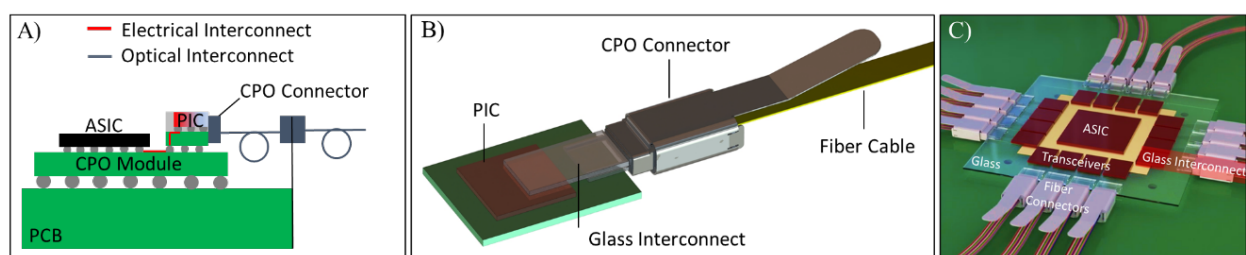


Fig. 1. A) CPO with detachable CPO connector. B) Detail of detachable CPO fiber connector. C) 16 flip-chip attached transceivers co-packaged with an ASIC on a glass substrate containing integrated glass interconnects providing a detachable fiber cable interface.

2. Glass Waveguide Interconnect

Integrated waveguides in glass are mode-matched to optical fibers and offer a compact and efficient solution for interconnecting PIC and fibers. The glass waveguides are made using a thermal ion-exchange (IOX) process on a 150 mm diameter alkali-glass wafer which is scalable to panel format and batch handling for cost reduction. The waveguide fabrication comprises a thin-film lithographic mask process to transfer the waveguide layout to the wafer, followed by a thermal ion-exchange phase to diffuse silver ions into the alkali-glass, increasing the refractive

index to form the optical waveguide core. After a second lithography process, the diffusion mask is removed, and critical alignment fiducials remain. A second diffusion step forms symmetric waveguides matching the single-mode fiber mode closely. The processed waveguide wafer is laser singulated with optical end-face quality and high-precision passive alignment trenches are made by ps-laser ablation for connector guide pin assembly [5]. The glass interconnect has an array of sixteen straight IOX waveguides with core pitch of 250 μm and length of 12.5 mm. The total measured insertion loss is 0.8 dB for coupling on both sides with a circular mode field diameter of 9.2 μm (e.g., single-mode fiber). Each interface has ~ 0.3 dB edge-coupling loss and the propagation loss is ~ 0.1 dB/cm at wavelength of 1310 nm.

A fan-out design shown in Figure 2A converts the 250 μm pitch connector interface to a 50 μm pitch for higher PIC density connectivity. The optical routing harnesses low-loss S-bend waveguides shown in Figure 2A, with the resulting simulated guided modes shown in Figure 2B. The S-bend is defined for a length of 9 mm and height of up to 1.5 mm for the outermost channels. This results in a minimum bend radius of 13.7 mm. The total device area is 12.5 mm $\times$ 6.5 mm. The simulated and experimental losses of the low-loss fan-out device are shown in Figure 2C and 2D where the performance of a standard fan-out is included for comparison. The bend radiation coefficient of the low-loss design at the minimum bend radius is estimated to be 0.015 dB/cm and 0.019 dB/cm for TE and TM polarizations, respectively. The total insertion loss (fiber-to-fiber) yields an average of 0.86 ± 0.13 dB, comprising of an estimated 0.3 dB of fiber coupling loss per facet, 0.126 dB of total propagation loss, and 0.14 ± 0.13 dB of bend loss.

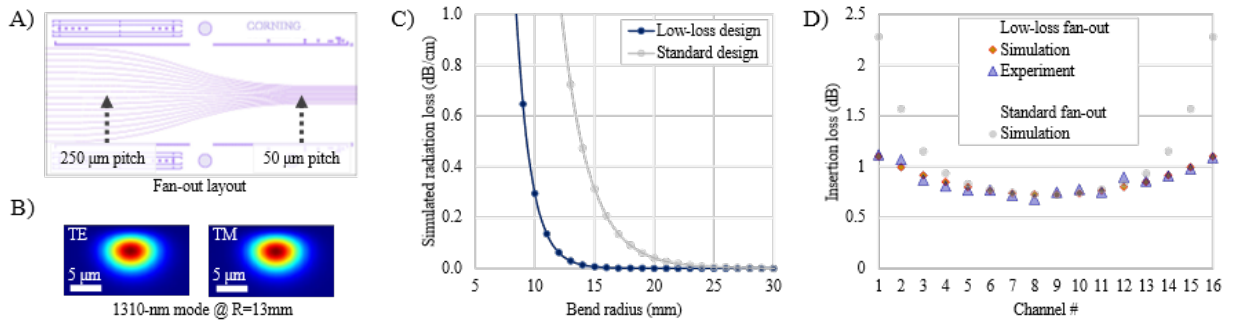


Fig. 2. Design and performance of a low-loss fan-out glass waveguide interconnect. A) Layout of the fanout depicting routing from 250 μm pitch down to 50 μm pitch. B) Modeled fanout waveguide showing the optical mode at a wavelength of 1310 nm and a bend radius of 13.7 mm. C) Simulated bend radiation loss (average of TE and TM) at varying bend radii and a standard non-optimized design for comparison. D) Total insertion loss (fiber-to-fiber) of the 16 channels in the low-loss fanout device; both simulation and experimental results are shown, and a reference standard fan-out is included for comparison.

3. Evanescent PIC Coupling and Detachable Fiber Connector

The connection between the glass interconnect and the PIC is done by evanescent coupling. As a proof of concept, a SiN PIC is flip-chip bonded to the fan-out waveguides shown in Figure 3A. The SiN PIC is composed of connected parallel waveguides forming nested U-loops (see Figure 3C inset) designed with a pitch of 100 μm to avoid bending losses. Linear tapers are employed to allow adiabatic evanescent coupling between the two platforms in the O-band. The SiN waveguides are 350 nm-thick, 3000 μm long, 700 nm wide and tapered to 260 nm. The IOX glass waveguides and the SiN waveguides are separated by an index-matching adhesive with 1 μm measured bondline. Demonstrations of low-loss coupling in the C-band have been previously described in [6]. Figure 3B shows the on-chip loss per facet in six devices for the TE and TM polarizations at 1310 nm. The results include bending and propagation losses of the IOX fan-out waveguides, and the evanescent coupling loss of IOX to SiN PIC. Edge coupling loss of the fiber array unit (FAU) to the IOX is not included. Figure 3C shows examples of the measured coupling loss per facet for the Device 1 (Blue) and Device 4 (Red) at both TE (solid lines) and TM (dashed lines) polarizations as a function of the wavelength, ranging from 1260 nm to 1360 nm. The Device 4 shows minimum loss of 0.38 dB at 1337 nm, and Device 1 shows losses below 1.65 dB in an extremely broad band range covering the entire O-band.

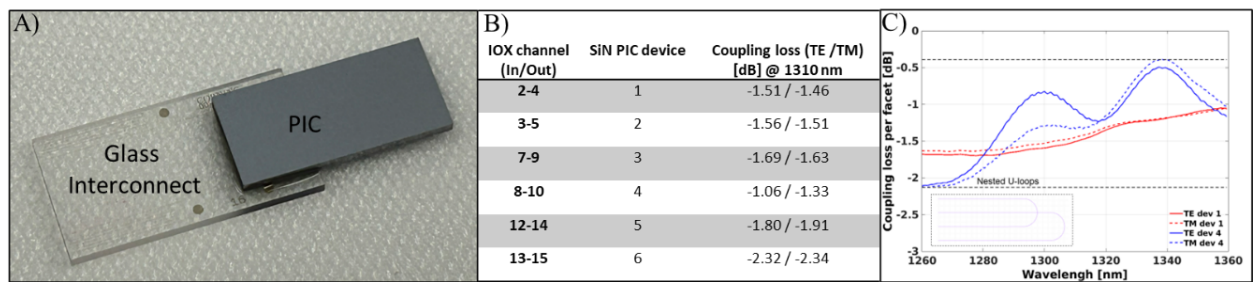


Fig. 3. A) Top view to PIC (5 mm width) assembled to glass interconnect (6.5 mm width). B) Coupling loss per facet at a wavelength of 1310 nm for different IOX channels. C) Measured coupling loss per facet as function of wavelength are shown for Device 1 (red) and Device (4) for both TE (solid line) and TM (dashed line) polarizations. The coupling loss per facet in B) and C) includes bending and propagation losses in the IOX waveguides, and the evanescent coupling loss of IOX to SIN PIC waveguide at the O-band. Not included is the edge coupling loss of FAU to IOX waveguide. The inset shows an illustration of the nested U-loops SiN structures.

Mechanical Transfer (MT) guide pins with a diameter of 550 μm are passively inserted into the alignment features and permanently bonded with a glass lid using adhesive bonding as shown in Figure 4A. The guide pins have grooves close to the tip providing a latching feature for mating a low-profile connector with physical contact, as shown in Figure 4B. This arrangement demonstrates a total interface loss of 0.8 dB including connector and mode-mismatch losses between fiber and IOX waveguides [7].

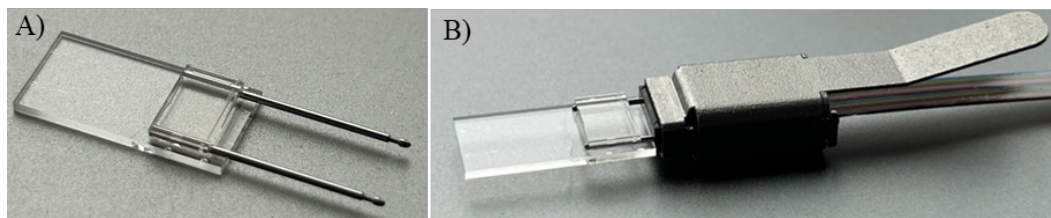


Fig. 4. A) Glass interconnect with two assembled 550 μm diameter guide pins for mating with a sixteen fiber MT ferrule connector. B) Mated low-profile fiber connector for fiber-to-glass waveguide edge-coupling.

4. Summary

Silicon photonics integrated circuits significantly contribute to the advancement of CPO due to their compatibility with the existing CMOS fabrication processes, allowing for the integration of optical and electronic components on a single chip, reducing costs, and encouraging high-volume manufacturing and wafer-level packaging. Both low loss evanescent chip coupling (min. 0.38 dB) and detachable fiber solutions (0.8 dB) are demonstrated by glass transfer waveguides with 0.1 dB/cm propagation loss for straight waveguides and 0.02 dB/cm bend radiation loss for 13.7 mm radius bent waveguides. Additional work is ongoing to combine both interfaces and demonstrate the full low-loss glass interconnect solution.

5. References

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